

# VISUAL ODOMETRY

Report submitted to GITAM (Deemed to be University) as a partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in (ECE)

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## **DECLARATION**

I/We declare that the project work contained in this report is original and it has been done by me under the guidance of my project guide.

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**CERTIFICATE**

This is to certify that **Thippeswamy ks** bearing **BU22EECE0100422** has satisfactorily completed Mini Project Entitled in partial fulfillment of the requirements as prescribed by University for VIIth semester, Bachelor of Technology in “Electrical, Electronics and Communication Engineering” and submitted this report during the academic year 2025-2026.

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## Chapter 1: Introduction

### 1.1 Overview of the Problem Statement

Accurate and reliable localization is a cornerstone for modern autonomous systems, including self-driving vehicles, mobile robots, and augmented reality applications. Traditionally, Global Positioning System (GPS) has served as the primary technology for determining global position. However, GPS-based systems suffer from significant limitations, particularly in environments where satellite signals are obstructed, attenuated, or unavailable. Such "GPS-denied" environments include dense urban canyons, tunnels, multi-story parking garages, forested areas, and indoor facilities. Furthermore, even when available, GPS can be susceptible to multi-path errors, spoofing, and varying degrees of accuracy, which are often insufficient for precise applications like lane-level navigation.

Alternative localization technologies, such as LiDAR or expensive Inertial Measurement Unit (IMU) arrays, offer high precision but come with substantial costs, computational overhead, and size/power constraints, limiting their widespread adoption, especially in cost-sensitive applications or smaller platforms. Stereo vision systems improve depth perception over monocular setups but double the computational load and sensor complexity.

This project addresses the critical need for a **low-cost, GPS-independent, and robust localization solution** that can operate effectively in these challenging environments. The core problem is to accurately and continuously determine a vehicle's 6-Degree-of-Freedom (6-DoF) pose (position and orientation) using only a monocular camera, while simultaneously mitigating the inherent drift associated with pure visual

odometry and achieving global consistency without reliance on external global positioning signals. The challenge lies in harmonizing continuous relative motion estimation with absolute position corrections derived from sparse, yet globally referenced, semantic features within a digital map.

## 1.2 Objectives and Goals

### 1.2.1 Objective

To develop and validate a **GPS-independent, monocular camera-based localization and path reconstruction system** that accurately determines a vehicle's global position and orientation by integrating Visual Odometry (VO) with real-time semantic feature matching against digital map data (e.g., OpenStreetMap). The system aims to provide a low-cost, robust, and deployable solution for navigation in GPS-denied or degraded environments.

### 1.2.2 Goals

The primary goals of the VisualMap Mono project are to implement robust visual odometry, develop a comprehensive semantic feature extraction pipeline, establish an efficient map data management and lookup system, design and implement an accurate pattern matching and localization module, achieve reliable path reconstruction and drift mitigation, and thoroughly evaluate the system's robustness and performance.

Specifically, the project aims to:

- **Implement Robust Visual Odometry (VO):**
  - Accurately estimate the vehicle's ego-motion (relative pose) frame-to-frame using a monocular camera.
  - Provide a continuous, albeit drift-prone, trajectory estimate as the foundational layer for localization.
- **Develop a Comprehensive Semantic Feature Extraction Pipeline:**

- Reliably detect, localize, and characterize diverse semantic cues from the camera feed (e.g., various signboards, road infrastructure, natural features).
- Extract actionable information such as text (via OCR) and geometric layouts from these detected features for map matching.
- **Establish an Efficient Map Data Management & Lookup System:**
  - Pre-load, store, and efficiently manage relevant vector and Point of Interest (POI) data from digital maps (e.g., OpenStreetMap).
  - Enable fast and contextual querying of the map database to retrieve potential matching features based on the current estimated vehicle pose.
- **Design and Implement an Accurate Pattern Matching & Localization Module:**
  - Effectively compare real-world semantic feature descriptors with corresponding digital map features using robust matching algorithms.
  - Integrate Visual Odometry estimates with high-confidence map matches via probabilistic filtering to produce a refined and globally consistent localization (position and orientation) and continuously correct Visual Odometry drift.
- **Achieve Reliable Path Reconstruction and Drift Mitigation:**
  - Construct a smooth and globally optimized vehicle trajectory by synthesizing map-corrected pose estimates over time.
  - Implement mechanisms for drift monitoring and intelligent re-localization triggers to ensure system stability and recovery, even after temporary periods of uncertainty or feature loss.
- **Evaluate System Robustness and Performance:**
  - Conduct rigorous testing to demonstrate accurate localization across diverse environmental conditions (e.g., varied lighting, weather) and complex infrastructure (e.g., intricate road networks, varying feature densities).
  - Assess computational efficiency, memory usage, and real-time

processing capabilities on target embedded hardware to confirm practical deployability.



## **Chapter 2 : Literature Review**

### **2.1 Introduction to Visual Odometry and SLAM**

Visual Odometry (VO) and Simultaneous Localization and Mapping (SLAM) are critical technologies enabling autonomous systems to navigate unknown environments. VO focuses on estimating the ego-motion of a camera by analyzing successive images, providing a relative pose estimation. SLAM, on the other hand, aims to simultaneously estimate the camera's pose and build a map of the environment. While traditional geometric approaches have dominated these fields for decades, recent advancements in deep learning have introduced powerful new paradigms. This chapter reviews key literature that underpins the VisualMap Mono project, encompassing both classical and learning-based methods, and highlighting their contributions to robust visual localization.

### **2.2 Foundations in Feature-Based SLAM**

The development of robust and versatile feature-based SLAM systems has been instrumental in advancing autonomous navigation. These methods rely on detecting and tracking salient features (e.g., corners, blobs) across image frames to estimate camera motion and build a consistent map.

- **ORB-SLAM: A Versatile and Accurate Monocular SLAM System** (Mur-Artal, Montiel, Tardós, IEEE Trans. Robotics 2015) ORB-SLAM is a landmark in feature-based SLAM, particularly for monocular setups. It introduced ORB features for robust feature detection and matching, combined with a sophisticated three-thread architecture (tracking, local mapping, loop closing) to achieve real-time performance and accuracy in diverse environments. Its emphasis on loop closure for global consistency is a key inspiration for mitigating drift, a central challenge in VisualMap Mono.
- **ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multi-Map SLAM** (Campos et al., IEEE

Trans. Robotics 2021) Building upon its predecessor, ORB-SLAM3 significantly extends the capabilities by incorporating visual-inertial fusion and multi-map capabilities. While VisualMap Mono is primarily vision-only, ORB-SLAM3's advancements in managing multiple, disconnected maps and achieving higher accuracy through sensor fusion provide valuable insights into robust state estimation and system scalability, especially for handling re-localization in previously mapped but temporarily lost regions.

- **Monocular Visual SLAM, Visual Odometry, and Structure from Motion Methods Applied to 3D Reconstruction (Survey Paper)** This survey paper provides a foundational understanding of the interconnected fields of VO, vSLAM, and Structure from Motion (SfM). It details the mathematical and algorithmic underpinnings of how 3D scene structure and camera motion can be reconstructed from 2D image sequences. For VisualMap Mono, this background is crucial for understanding the limitations of monocular depth estimation and scale ambiguity, which our map-matching approach aims to resolve.

### 2.3 Evolution and Application of Visual Odometry

Visual Odometry forms the real-time motion estimation core of SLAM systems. Its evolution has seen significant strides from purely geometric to data-driven learning-based approaches.

- **Evolution of Visual Odometry Techniques (Review Article)** This comprehensive review article traces the development of VO from its early geometric formulations to more recent learning-based paradigms. It covers various aspects, including feature-based vs. direct methods, monocular vs. stereo/RGB-D setups, and the inherent challenges like drift. For VisualMap Mono, this review helps contextualize the choice of a monocular setup and validates the necessity of external global

corrections due to VO's drift characteristics.

- **Feature-Based Monocular Visual Odometry for Autonomous Driving and Hyperparameter Tuning to Improve Trajectory Estimation (Applied Research Paper)** This applied research paper directly addresses challenges relevant to VisualMap Mono in an autonomous driving context. It focuses on optimizing feature-based monocular VO through hyperparameter tuning to enhance trajectory estimation accuracy. Its insights are valuable for refining the initial VO module in VisualMap Mono, ensuring a more stable and accurate baseline motion estimate before map correction.
- **Understanding Visual SLAM for Robotics Perception: Building Monocular SLAM from Scratch in Python OpenCV (Educational Resource / Tutorial)** This educational resource provides practical guidance on implementing monocular vSLAM pipelines. While not a research paper, it offers invaluable hands-on insights into the practical aspects of feature detection, tracking, and pose estimation using libraries like OpenCV. This resource directly informs the initial development and prototyping stages of the Visual Odometry and Image Pre-processing modules in VisualMap Mono.

#### 2.4 Emergence of Deep Learning in Visual Localization

Deep learning has revolutionized various computer vision tasks, including feature extraction, depth estimation, and direct pose regression, leading to more robust and generalized VO and SLAM systems.

- **Deep Direct Visual Odometry (Zhou et al., CVPR 2017)** This pioneering work introduced an unsupervised deep learning approach for monocular VO, moving away from explicit feature detection. By training a convolutional neural network to predict camera motion and depth from raw image pixels, it demonstrated the potential of direct methods combined with learning to handle texture-less

areas and illumination changes. VisualMap Mono can draw upon this work for potentially robustifying its VO module or enhancing semantic feature extraction beyond traditional methods.

- **DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras (Zhongjie Yu et al., NeurIPS 2021)**  
DROID-SLAM represents a significant advancement in deep learning-based SLAM, offering a unified framework for various camera types. Its use of recurrent feature alignment and optimization-based pose estimation addresses challenges like severe motion and dynamic environments. For VisualMap Mono, DROID-SLAM offers a benchmark for the level of robustness and accuracy achievable with deep learning, and its architectural concepts could inspire future enhancements for more robust feature tracking and pose graph optimization.
- **SpatialTrackerV2: 3D Point Tracking Made Easy (Xu et al., CVPR 2024)**  
This recent learning-based method for dense 3D point tracking is highly relevant for robustifying the feature tracking and pose estimation components of SLAM/VO systems. By providing dense and consistent 3D correspondences, SpatialTrackerV2 could significantly improve the accuracy and stability of VisualMap Mono's Visual Odometry module, especially in environments where sparse features are less reliable. Its deep learning backbone can lead to more generalized feature handling than traditional hand-crafted methods.

## **Chapter 3 : Strategic Analysis and Problem Definition**

### 3.1 SWOT Analysis

### 3.2 Project Plan - GANTT Chart

### 3.3 Problem statement

## Chapter 4: Methodology

### 4.1 Description of the Approach

The VisualMap Mono project employs a hybrid, modular approach to achieve GPS-independent localization and path reconstruction. The core idea is to combine the continuous relative motion estimation capability of Visual Odometry (VO) with the global consistency provided by semantic feature matching against a digital map. This strategy allows the system to leverage the strengths of both paradigms while mitigating their individual weaknesses, primarily VO drift and the need for persistent, unique map features for re-localization.

The system workflow can be broadly categorized into the following sequential and interlinked steps:

1. **Initialization:** The system starts with a user-provided rough initial geographical coordinate (latitude, longitude). Based on this, relevant map data (road networks, POIs, semantic features) are pre-loaded from OpenStreetMap.
2. **Visual Odometry (VO):** As the vehicle moves, the monocular camera continuously captures images. These images are processed to estimate the camera's pose (position and orientation) frame-by-frame. This generates a continuous local trajectory but is inherently subject to accumulation of errors, leading to drift over longer distances.
3. **Semantic Feature Extraction:** In parallel with VO, pre-processed images are analyzed to detect and extract semantic features from the environment. These features are not just generic visual points but semantically meaningful objects like specific road signs (e.g., "STOP," "Ring Road"), road types (e.g., "National Highway"), junctions, and other distinctive landmarks.
4. **Feature Description and Map Pre-processing:** Detected semantic features are transformed into robust descriptors that

can be compared against map data. Concurrently, the pre-loaded map data is processed to create a database of geo-referenced semantic features and their associated descriptors or attributes (e.g., signboard text, junction topology).

5. **Pattern Matching and Localization:** The current estimated vehicle pose from VO provides an approximate search area on the map. Within this area, the extracted real-world semantic features (and their descriptors) are compared against the geo-referenced semantic features from the map database. High-confidence matches enable the system to calculate an absolute global pose correction.
6. **Pose Fusion and Drift Mitigation:** The drift-prone VO pose estimates are fused with the absolute pose corrections derived from map matching using probabilistic filtering techniques (e.g., a Kalman filter or particle filter). This fusion process continuously refines the vehicle's estimated global position and orientation, effectively mitigating VO drift.
7. **Path Reconstruction and Trajectory Optimization:** The sequence of corrected global poses is used to build a smooth, consistent, and globally optimized trajectory. Mechanisms for drift monitoring and re-localization triggers ensure that if localization confidence drops, the system attempts to re-establish a strong map match.

#### 4.2 Tools and Techniques Utilized

The implementation of VisualMap Mono will leverage a combination of established computer vision libraries, deep learning frameworks, and geospatial tools:

- **Programming Language:** Python will be the primary language for development due to its extensive libraries for computer vision, deep learning, and data processing. Critical performance-intensive modules may use C++ for optimization.



- Computer Vision Libraries:
  - OpenCV: For fundamental image processing tasks (e.g., camera calibration, distortion correction, image normalization), traditional feature detection and tracking (e.g., ORB, SIFT, optical flow for VO), and geometric transformations.
- Deep Learning Frameworks:
  - TensorFlow / PyTorch: For implementing and training deep learning models used in semantic feature extraction (object detection, semantic segmentation, OCR) and potentially for learning-based VO components.
- Geospatial Libraries:
  - OSMnx / GEOS (via Shapely/Fiona): For parsing, processing, querying, and visualizing OpenStreetMap data, including road networks, POIs, and geographic coordinates.
- Visual Odometry / SLAM Frameworks (for baseline/comparison):
  - Potentially use components or insights from open-source VO/SLAM projects (e.g., ORB-SLAM's feature handling, direct methods for pose estimation) for robust VO implementation.
- Probabilistic Filtering:
  - Custom implementations of Extended Kalman Filters (EKF) or Particle Filters for fusing VO estimates with map match corrections.



## 4.1 Design considerations

Several key design considerations guide the development of VisualMap Mono to ensure its robustness, accuracy, and practical deployability:

- **Modularity:** The system is designed with distinct, interchangeable modules (Image Pre-processing, VO, Feature Extraction, Map Matching, Pose Fusion). This facilitates independent development, testing, and future upgrades (e.g., swapping out a traditional VO module for a deep learning-based one).
- **Scalability of Map Data:** The map data management system is designed to efficiently handle large geographical areas. This includes tiling strategies, hierarchical data structures, and optimized querying to ensure real-time access to relevant map features.
- **Robustness to Environmental Variations:** Algorithms for image pre-processing, feature extraction, and matching are designed with resilience to varying lighting (day/night, shadows), weather (rain, fog, glare), and seasonal changes (vegetation, construction) in mind. This may involve adaptive thresholds, normalization techniques, and deep learning models trained on diverse datasets.
- **Drift Mitigation:** The fusion of VO with absolute map matches is central. The choice of probabilistic filter and its tuning will be critical to effectively bound VO drift and provide a globally consistent pose estimate. Mechanisms for monitoring localization confidence will trigger re-localization attempts if necessary.
- **Real-time Performance:** Given the target applications (e.g., autonomous driving), the system's computational load, memory usage, and latency are critical. Optimization techniques such as model pruning, quantization for deep learning models, and efficient algorithm implementations will be considered for deployment on embedded platforms (e.g., NVIDIA Jetson).

- **Semantic Feature Diversity:** The system is designed to leverage a wide array of semantic features, moving beyond generic points. This includes text on signs, geometric layouts of junctions, specific road markings, and even patterns in natural terrain, enhancing the uniqueness and density of matchable features.
- **Initialization and Re-localization:** The system needs robust procedures for initializing its position from a rough estimate and for efficiently re-localizing itself after periods of tracking loss or extended absence of map features.
- **Accuracy vs. Robustness Trade-offs:** Conscious decisions will be made regarding the balance between achieving high localization accuracy and maintaining robustness in challenging, real-world conditions, often prioritizing robustness for safety-critical applications.

## **Chapter 5 : Implementation**

5.1 Description of how the project was executed

5.2 Challenges faced and solutions implemented

## **Chapter 6: Results**

### 6.1 outcomes

### 6.2 Interpretation of results

### 6.3 Comparison with existing literature or technologies

## **Chapter 7: Conclusion**

Here write Suggestions for further research or development and Potential improvements or extensions

## **Chapter 8 : Future Work**

Here write Suggestions for further research or development  
Potential improvements or extensions

## References



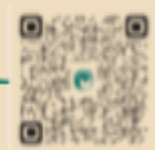
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